Customer Churn Prediction Project Report

Executive Summary

Customer churn is a critical challenge for banking institutions because acquiring new customers is significantly more expensive than retaining existing ones. This project aims to predict whether a customer is likely to leave the bank using historical customer data and machine learning techniques.

Several Machine Learning and Deep Learning models were developed and evaluated. After comparing performance metrics, XGBoost was selected as the final model due to its strong predictive performance and good generalization capability.

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1. Business Problem

Banks face revenue loss when customers close their accounts or move to competing institutions.

Business Objective

Develop a predictive model that can identify customers likely to churn so that the bank can take proactive retention actions.

Expected Business Benefits

- Improved customer retention
- Reduced customer acquisition costs
- Better targeted marketing campaigns
- Increased customer lifetime value

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2. Dataset Description

The dataset contains customer demographic and banking information.

Target Variable

Exited

- 1 = Customer Churned
- 0 = Customer Retained

Features Used

- Credit Score
- Geography
- Gender
- Age
- Tenure
- Balance

- Number of Products
- Credit Card Status
- Active Membership Status
- Estimated Salary

Removed Features

The following columns were removed because they do not contribute meaningful predictive information:

- RowNumber
- CustomerId
- Surname

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3. Data Preprocessing

The following preprocessing steps were performed:

Data Cleaning

- Missing value verification
- Duplicate record verification
- Data type validation
- Removal of irrelevant columns

Feature Engineering

Label Encoding

Gender:

- Male = 1
- Female = 0

One-Hot Encoding

Geography column converted into numerical features.

Data Splitting

Training Data: 80%

Testing Data: 20%

Feature Scaling

StandardScaler was applied for Neural Network models.

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4. Exploratory Data Analysis

EDA was conducted to understand customer behavior and identify important patterns.

Techniques Used

- Histogram Analysis
- Count Plot Analysis
- Box Plot Analysis
- Correlation Analysis
- Feature vs Target Analysis

Key Findings

- Older customers showed higher churn tendencies.
- Active members were less likely to churn.
- Geography had a noticeable impact on churn behavior.
- Customers with fewer products demonstrated higher churn risk.

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5. Model Development

The following models were developed and evaluated:

Model 1: Perceptron

Results:

- Accuracy: 81.2%
- Recall: 20.6%
- F1 Score: 30.0%
- ROC-AUC: 58.3%

Model 2: Multi-Layer Perceptron (MLP)

Results:

- Accuracy: 86.1%
- Recall: 47.3%
- F1 Score: 57.7%
- ROC-AUC: 71.6%

Model 3: Deep Neural Network (DNN)

Architecture:

Input Layer → 22 Neurons → 11 Neurons → Output Layer

Results:

- Accuracy: 86.3%
- Recall: 50.9%
- F1 Score: 59.3%
- ROC-AUC: 72.9%

Model 4: Decision Tree

Results:

- Accuracy: 86.0%
- Recall: 45.3%
- F1 Score: 56.0%
- ROC-AUC: 70.6%

Model 5: Random Forest

Results:

- Accuracy: 86.55%
- Recall: 46.8%
- F1 Score: 57.8%
- ROC-AUC: 71.5%

Model 6: XGBoost

Results:

- Accuracy: 86.7%
- Precision: 74%
- Recall: 50.1%
- F1 Score: 60.0%
- ROC-AUC: 72.9%

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6. Model Comparison

Model	Accuracy	Recall	F1 Score	ROC-AUC
Perceptron	81.2%	20.6%	30.0%	58.3%
MLP	86.1%	47.3%	57.7%	71.6%
DNN	86.3%	50.9%	59.3%	72.9%

Model	Accuracy	Recall	F1 Score	ROC-AUC
Decision Tree	86.0%	45.3%	56.0%	70.6%
Random Forest	86.55%	46.8%	57.8%	71.5%
XGBoost	86.7%	50.1%	60.0%	72.9%

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7. Final Model Selection

Selected Model

XGBoost

Reason for Selection

The XGBoost model achieved the best overall balance between:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

It also demonstrated excellent generalization performance with minimal overfitting.

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8. Business Recommendations

Based on the model findings:

1. Focus retention campaigns on customers identified as high-risk.
2. Improve engagement programs for inactive members.
3. Monitor customers with low product usage.
4. Design personalized offers for customers showing churn indicators.
5. Use the predictive model as part of the customer relationship management workflow.

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9. Conclusion

A complete end-to-end Customer Churn Prediction system was successfully developed using Machine Learning and Deep Learning techniques.

Among all evaluated models, XGBoost achieved the strongest overall performance and was selected as the final model. The solution can help the bank identify customers at risk of churn and support proactive retention strategies, ultimately reducing customer attrition and improving business profitability.